

CNC MACHINE COMPLETE KNOWLEDGE BASE

Industrial Reference Manual for IndusBot Backend

1. INTRODUCTION TO CNC MACHINES

What is a CNC Machine?

CNC stands for Computer Numerical Control.

A CNC machine is an automated manufacturing machine controlled by computer programs. It is used to perform machining operations such as:

- Cutting
- Drilling
- Milling
- Turning
- Boring
- Tapping
- Grinding
- Engraving

The machine follows programmed instructions called G-code and M-code to manufacture precise components.

CNC machines are widely used in:

- Automotive industries
 - Aerospace industries
 - Manufacturing industries
 - Medical industries
 - Tool rooms
 - Mold industries
 - Heavy engineering
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2. TYPES OF CNC MACHINES

2.1 CNC LATHE MACHINE

Purpose

Used for turning cylindrical components.

Main Operations

- Facing
- Turning
- Grooving
- Threading
- Drilling
- Boring
- Knurling

Axes

- X Axis
- Z Axis

Common Applications

- Shafts
 - Bushes
 - Rollers
 - Threaded parts
-

2.2 CNC MILLING MACHINE

Purpose

Used for cutting and shaping solid materials.

Main Operations

- Face milling
- End milling
- Slot milling
- Pocketing
- Drilling

Axes

- X Axis
 - Y Axis
 - Z Axis
-

2.3 VMC (VERTICAL MACHINING CENTER)

Definition

A CNC machine with vertical spindle orientation.

Features

- Automatic Tool Changer (ATC)
- High precision
- Multiple tool operations

Applications

- Mold manufacturing
 - Automotive parts
 - Precision machining
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2.4 HMC (HORIZONTAL MACHINING CENTER)

Definition

A CNC machine with horizontal spindle orientation.

Advantages

- Better chip removal
 - Higher productivity
 - Suitable for heavy cutting
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2.5 CNC ROUTER

Used For

- Wood cutting
 - Acrylic cutting
 - Plastic cutting
 - Aluminum cutting
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2.6 5-AXIS CNC MACHINE

Features

- Simultaneous multi-axis movement
- Complex machining capability
- High precision

Industries

- Aerospace
 - Medical implants
 - Turbine manufacturing
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3. MAIN COMPONENTS OF CNC MACHINE

3.1 MACHINE BED

Function

Provides structural support.

Material

Usually made of cast iron.

Importance

- Absorbs vibration
 - Maintains machine rigidity
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3.2 SPINDLE

Function

Rotates cutting tool or workpiece.

Types

- Belt-driven spindle
- Direct-drive spindle
- Motorized spindle

Common Issues

- Overheating
 - Vibration
 - Bearing failure
-

3.3 AXES SYSTEM

Linear Axes

- X Axis
- Y Axis
- Z Axis

Rotary Axes

- A Axis
- B Axis

- C Axis

Axis Direction

- X → Left/Right
 - Y → Front/Back
 - Z → Up/Down
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3.4 TOOL MAGAZINE

Purpose

Stores cutting tools.

Capacity

Can store multiple tools.

Types

- Drum type
 - Chain type
 - Carousel type
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3.5 AUTOMATIC TOOL CHANGER (ATC)

Function

Automatically changes tools.

Benefits

- Reduces cycle time
- Improves productivity

Common Problems

- Tool clamp failure
 - Tool position mismatch
 - ATC arm jam
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3.6 CHUCK

Function

Holds workpiece.

Types

- Hydraulic chuck
 - Pneumatic chuck
 - Manual chuck
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3.7 SERVO MOTOR

Function

Controls precise axis movement.

Advantages

- High accuracy
 - Fast response
 - Controlled positioning
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3.8 BALL SCREW

Function

Converts rotational motion into linear motion.

Advantages

- High precision
 - Low friction
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3.9 GUIDEWAYS

Purpose

Guides axis movement.

Types

- Box guideways
 - Linear guideways
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3.10 CONTROL PANEL

Purpose

Operator interface for machine control.

Contains

- Display screen
 - Emergency stop
 - Feed controls
 - Program buttons
 - Manual controls
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4. CNC CONTROLLER SYSTEM

Common CNC Controllers

FANUC

Widely used industrial controller.

SIEMENS

Advanced CNC automation system.

MITSUBISHI

Used in industrial machining centers.

HAAS

Popular in American CNC machines.

FAGOR

Used in European CNC systems.

5. CNC MACHINE AXIS EXPLANATION

X AXIS

Horizontal movement.

Y AXIS

Depth movement.

Z AXIS

Vertical movement.

A/B/C AXIS

Rotational axes.

6. MACHINE STARTUP PROCEDURE

Step-by-Step Startup

1. Switch ON main power supply.
 2. Release Emergency Stop.
 3. Start hydraulic system.
 4. Start lubrication system.
 5. Reference return (Home position).
 6. Check air pressure.
 7. Warm up spindle.
 8. Load program.
 9. Set tool offsets.
 10. Set work offsets.
 11. Run dry cycle.
 12. Start machining.
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7. MACHINE SHUTDOWN PROCEDURE

1. Stop machining cycle.
 2. Move axes to safe position.
 3. Stop spindle.
 4. Remove workpiece.
 5. Clean machine.
 6. Switch OFF hydraulics.
 7. Switch OFF lubrication.
 8. Turn OFF machine power.
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8. MACHINE HOMING

What is Homing?

Machine moves to reference position.

Purpose

- Establishes machine coordinates
- Initializes axis position

Common Homing Problems

- Axis overtravel
 - Encoder error
 - Limit switch fault
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9. TOOL OFFSET

Definition

Distance compensation between tool tip and machine reference.

Types

- Length offset
- Radius offset

Purpose

- Accurate machining
 - Tool compensation
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10. WORK OFFSET

Definition

Defines workpiece zero point.

Common Offsets

- G54
 - G55
 - G56
 - G57
 - G58
 - G59
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11. G-CODES

What are G-Codes?

Programming commands controlling machine movement.

Common G-Codes

G-Code	Function
G00	Rapid positioning
G01	Linear interpolation
G02	Circular interpolation clockwise
G03	Circular interpolation counterclockwise
G04	Dwell
G17	XY plane selection
G18	ZX plane selection
G19	YZ plane selection
G20	Inch programming
G21	Metric programming
G28	Return to machine home
G40	Cutter compensation cancel
G41	Cutter compensation left
G42	Cutter compensation right
G43	Tool length compensation positive
G49	Tool compensation cancel
G54-G59	Work coordinate systems
G73	High-speed drilling cycle
G81	Drilling cycle
G83	Peck drilling cycle
G90	Absolute programming
G91	Incremental programming
G94	Feed per minute
G95	Feed per revolution

12. M-CODES

What are M-Codes?

Machine function control commands.

Common M-Codes

M-Code	Function
M00	Program stop
M01	Optional stop
M03	Spindle clockwise
M04	Spindle counterclockwise
M05	Spindle stop
M06	Tool change
M08	Coolant ON
M09	Coolant OFF
M19	Spindle orientation
M30	Program end and reset
M98	Subprogram call
M99	Subprogram return

13. SAMPLE CNC PROGRAM

Facing Program Example

O1001; G21; G90; T0101; M06; G00 X100 Z5; M03 S1200; G01 Z0 F0.2; G01 X0; G00 Z5; M05; M30;

14. CUTTING TOOLS

Common Tool Types

End Mill

Used for milling operations.

Drill

Used for hole making.

Boring Tool

Used for enlarging holes.

Threading Tool

Used for thread cutting.

Grooving Tool

Used for grooves.

15. TOOL MATERIALS

Material	Application
HSS	General machining
Carbide	High-speed cutting
Ceramic	Hard materials
CBN	Hardened steel
Diamond	Non-ferrous materials

16. COOLANT SYSTEM

Purpose

- Reduce heat
- Improve tool life
- Remove chips

Types

- Flood coolant
- Mist coolant
- Air cooling

Common Problems

- Low coolant level
 - Pump failure
 - Blocked nozzle
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17. LUBRICATION SYSTEM

Purpose

Lubricates moving components.

Components

- Lubrication pump
- Oil reservoir
- Distribution lines

Common Issues

- Low oil level
 - Pump malfunction
 - Blocked lines
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18. HYDRAULIC SYSTEM

Purpose

Controls:

- Chuck clamping
- Tool clamping
- ATC operations

Common Problems

- Low pressure
 - Oil leakage
 - Pump failure
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19. PNEUMATIC SYSTEM

Uses

- Air blast
- Tool unclamping
- Pneumatic actuators

Required Pressure

Usually 5–7 bar.

Common Issues

- Air leakage
 - Low pressure
 - Moisture in airlines
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20. CNC MACHINE ALARMS

Common Alarm Categories

Servo Alarm

Axis movement issue.

Spindle Alarm

Spindle malfunction.

Overtravel Alarm

Axis exceeded limit.

Hydraulic Alarm

Hydraulic pressure issue.

Lubrication Alarm

Low lubrication.

ATC Alarm

Tool changer issue.

Encoder Alarm

Position feedback problem.

21. COMMON CNC MACHINE PROBLEMS

Problem: Spindle Not Rotating

Possible Causes

- Spindle drive fault
- Emergency stop active
- Overload condition
- Motor failure

Solutions

- Reset alarm
 - Check spindle drive
 - Check power supply
 - Inspect motor
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Problem: Machine Not Homing

Causes

- Sensor fault
- Encoder issue
- Axis obstruction

Solutions

- Check limit switches
 - Inspect encoder
 - Remove obstruction
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Problem: Poor Surface Finish

Causes

- Tool wear
- Vibration
- Incorrect speed/feed

Solutions

- Replace tool
 - Optimize cutting parameters
 - Check spindle rigidity
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Problem: Tool Breakage

Causes

- Excessive feed
- Wrong tool selection
- Poor coolant flow

Solutions

- Reduce feed
 - Use proper tool
 - Improve coolant supply
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22. CNC MACHINE MAINTENANCE

Daily Maintenance

- Clean machine
 - Check coolant level
 - Check lubrication level
 - Remove chips
 - Inspect air pressure
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Weekly Maintenance

- Inspect tool holder
 - Clean filters
 - Check hydraulic pressure
 - Inspect belts
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Monthly Maintenance

- Check spindle alignment
 - Inspect ball screws
 - Check backlash
 - Verify axis calibration
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Yearly Maintenance

- Complete machine inspection
- Servo tuning
- Replace worn components
- Electrical inspection

23. SAFETY PRECAUTIONS

General Safety Rules

1. Wear safety glasses.
 2. Do not touch rotating spindle.
 3. Keep doors closed during machining.
 4. Use proper PPE.
 5. Avoid loose clothing.
 6. Follow lockout procedures.
 7. Do not bypass safety interlocks.
 8. Stop machine during inspection.
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24. EMERGENCY STOP SYSTEM

Purpose

Immediately stops machine operation.

When to Use

- Collision
 - Tool breakage
 - Smoke/fire
 - Unsafe conditions
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25. CNC MACHINE PARAMETERS

Definition

Machine configuration settings.

Examples

- Axis travel limits
 - Servo tuning
 - Spindle settings
 - Feed limits
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26. CNC MACHINE ACCURACY

Types of Accuracy

Positioning Accuracy

How accurately machine reaches commanded position.

Repeatability

Ability to return to same position repeatedly.

27. CNC MACHINE CALIBRATION

Purpose

Maintain machining accuracy.

Calibration Includes

- Axis alignment
 - Tool calibration
 - Probe calibration
 - Spindle alignment
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28. INDUSTRIAL APPLICATIONS

Automotive Industry

Engine components, shafts, gears.

Aerospace Industry

Turbine blades, aircraft components.

Medical Industry

Implants, surgical tools.

Mold Industry

Injection molds, dies.

29. CNC MACHINE ADVANTAGES

1. High precision
 2. Repeatability
 3. Reduced human error
 4. Faster production
 5. Complex machining capability
 6. High productivity
 7. Automation support
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30. CNC MACHINE LIMITATIONS

1. High initial cost
 2. Skilled programming required
 3. Maintenance cost
 4. Dependency on electrical systems
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31. INDUSTRY 4.0 AND CNC

Modern CNC Features

- IoT integration
 - Smart monitoring
 - Predictive maintenance
 - Remote diagnostics
 - AI integration
 - Digital twin systems
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32. COMMON OPERATOR QUESTIONS FOR INDUSBOT

Machine Operation Queries

- How to start CNC machine?
- Why spindle not rotating?
- How to reset alarm?
- What is G54?
- How to set tool offset?
- Why hydraulic pressure low?
- How to home machine?
- Why ATC not working?
- How to set work zero?

- Why machine vibration occurring?
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33. SAMPLE TROUBLESHOOTING DATA FOR INDUSBOT

Alarm: LOW LUBRICATION PRESSURE

Possible Causes

- Low oil level
- Lubrication pump failure
- Leakage in lubrication line

Recommended Actions

1. Check lubrication oil level.
 2. Inspect lubrication pump.
 3. Verify pressure sensor.
 4. Check for oil leakage.
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Alarm: SPINDLE OVERLOAD

Causes

- Heavy cutting load
- Tool jam
- Bearing issue

Solutions

1. Reduce feed rate.
 2. Check cutting tool.
 3. Inspect spindle bearings.
 4. Restart spindle.
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34. CNC MACHINE TERMINOLOGIES

Term	Meaning
RPM	Revolutions per minute
Feed Rate	Tool movement speed
DOC	Depth of cut

Term	Meaning
WCS	Work coordinate system
ATC	Automatic tool changer
HMI	Human machine interface
PLC	Programmable logic controller
Servo	Precision motion motor
Backlash	Mechanical play

35. BEST PRACTICES

1. Always perform dry run.
2. Check offsets before machining.
3. Use proper cutting parameters.
4. Maintain coolant level.
5. Regularly inspect tools.
6. Clean machine daily.
7. Monitor unusual sounds.
8. Follow preventive maintenance.

36. FUTURE OF CNC MACHINES

Future Technologies

- AI-powered machining
- Autonomous CNC systems
- Smart factories
- Predictive analytics
- Cloud-connected CNC
- Robotic integration
- Digital manufacturing

END OF CNC MACHINE KNOWLEDGE BASE